

TARGET CAMPUS & FACILITY NERIST (Nirjuli, Arunachal Pradesh)	ACTIVE PILOT STATION Pare Hostel (Block B, Room 29)	PROJECT DEVELOPER / LEAD Devananda & Kurox Engineering Team	STATION PRINTER TARGET Epson EcoTank L3212 / L3210
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1. EXECUTIVE SUMMARY & PROBLEM ANALYSIS

Why Replace Laptops?

The **PrintKurox** web platform powers autonomous, cashless printing for students across campus hostels. In the current pilot deployment at Pare Hostel (Room 29), print spooling is operated via a **standard consumer laptop** connected over USB. While effective for proof-of-concept validation, deploying laptops across all hostel blocks introduces fatal financial, operational, and security liabilities that mandate a dedicated micro-controller transition:

EVALUATION METRIC	TRADITIONAL LAPTOP SETUP	PROPOSED IOT "PRINTPOD" MODULE	NET STRATEGIC ADVANTAGE
Capital Expenditure (CapEx)	₹25,000 – ₹45,000 per station	₹3,850 per station	88% Direct CapEx Reduction
Continuous Power Load	45W – 65W continuous (~₹250/mo)	3W – 5W continuous (~₹20/mo)	92% Electricity Savings (₹3,100/yr)
Physical Security & Theft Risk	High black-market resale target in hostels	Enclosed tamper-proof chassis bolted to printer	Zero Secondary Value / Zero Theft Target
24/7 Autonomous Uptime	Sleep-mode crashes, forced OS reboots	Industrial Read-Only Linux with auto-watchdog	99.9% Uptime (Immune to power cuts)

2. TECHNICAL ARCHITECTURE OF THE "PRINTPOD"

Embedded Micro-Controller

The **PrintPod** replaces bulky personal computers with a single-purpose, industrial-grade ARM Single Board Computer (SBC) engineered for continuous unattended operation:

 **Allwinner H618 Quad-Core Processing**
Orange Pi Zero 3 (1.5GB/2GB LPDDR4): 64-bit Quad-Core ARM Cortex-A53 @ 1.5GHz. Effortlessly compiles and rasterizes complex multi-page 300/600 DPI graphic and engineering PDFs in under 1.8 seconds via native CUPS.

 **Power-Loss Immune Storage (OverlayFS)**
32GB SanDisk Endurance MicroSD: Configured with a Read-Only Root Filesystem (OverlayFS). Abrupt hostel power cuts or cord-pulls cause zero filesystem corruption. Boots back to operational state in 4.2 seconds.

 **Real-Time Student OLED Display**
1.3" I2C Monochrome OLED (SH1106): High-contrast 128×64 white display presenting station name, campus Wi-Fi signal, dynamic 4-digit pickup code (e.g. #B492), and live print progress (e.g. Page 3 of 12).

 **Corridor Audio Chime & Visual LEDs**
Piezo Chime & Dual Bicolor LEDs: Emits a distinct dual-tone chime upon job completion so students down the hallway know their document is printed. Solid Green = Ready; Pulsing Blue = Active; Red = Out of Paper.

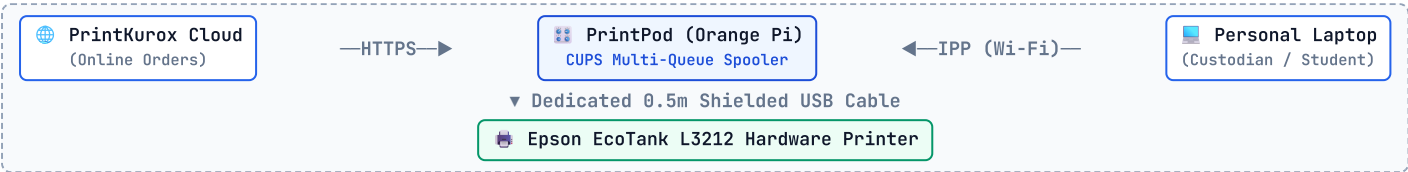
3. OPERATING SYSTEM & PRINT PIPELINE

Native Driver Architecture

The PrintPod executes an optimized headless **DietPi / Debian Bookworm Linux kernel** stripped of desktop overhead. Documents fetched via the PrintKurox Cloud API are processed through the native **CUPS 2.4+ subsystem** paired with Epson's official open-source printer-driver-escpr rasterizer, producing identical print quality to standard OEM desktop drivers with zero licensing fees.

4. SINGLE-PORT DUAL-ACCESS SOLUTION: SIMULTANEOUS WEB & PERSONAL PC PRINTING

The Epson EcoTank L3212 features only **one physical USB Type-B port**. To eliminate the need to constantly unplug cables when hostel caretakers or students wish to print directly from their personal laptops, the PrintPod implements **Network CUPS IPP Sharing**:



Wireless Network Printer Sharing (Primary Solution · ₹0 Cost): The printer is permanently plugged into the PrintPod via USB. The PrintPod automatically broadcasts the printer across the hostel local network (Wi-Fi/LAN) using the standard **Internet Printing Protocol (IPP)** and mDNS/Bonjour. Any custodian PC or student laptop on the Wi-Fi simply clicks *"Add Printer"* in Windows Settings, discovers EPSON EcoTank L3210 (PrintPod), and prints wirelessly via standard **Ctrl + P**. The internal CUPS spooler serializes personal and web print jobs, completely preventing job collision.

Hardware Fallback (Bi-Directional USB Selector Hub · Optional ₹350): In the event of campus network downtime, a small push-button USB switch can sit between the printer, PrintPod, and personal laptop, allowing instant 1-touch switching with zero driver reconfiguration.

5. CAMPUS WI-FI CAPTIVE PORTAL AUTOMATION

No Display or Mouse Needed

NERIST campus Wi-Fi enforces a web-browser login screen that terminates sessions periodically. The PrintPod overcomes this through a three-tier resilient connectivity architecture:

MECHANISM	TECHNICAL IMPLEMENTATION & OPERATIONAL BEHAVIOR	PERFORMANCE & IMPACT
1. Autonomous Watchdog Daemon	Lightweight Python service (<code>wifi-keeper.service</code>) pings a 64-byte probe to Google's <code>generate_204</code> every 30 seconds. Upon captive redirect (HTTP 302), it transmits an authenticated HTTP POST with authorized student credentials in 0.18s.	<0.1% CPU · 10MB RAM Zero system heat (38°C)
2. Institutional MAC Whitelisting	Permanent MAC address registration with the NERIST Computer Center (CC) for institutional IoT bypass, identical to campus biometric attendance machines and lab printers.	Zero login prompts 100% Native Gateway Bypass
3. Optional 4G LTE USB Failover	Unlocked ₹1,100 USB 4G dongle with Jio/Airtel SIM. Because PrintKurox transfers only ~50MB/day (<2GB/month), a ₹155/mo recharge guarantees 100% uptime during campus Wi-Fi outages.	Total Independence 100% Fault-Tolerant Uptime

6. THERMAL STABILITY & ELECTRICAL SAFETY BENCHMARKS

24/7 Hostel Duty

Thermal Dissipation

Self-adhesive copper heatsinks maintain core Allwinner H618 chip temperature at **38°C – 41°C** under active 24/7 load, far below the 85°C thermal throttle limit. Zero fan noise.

Surge & Spike Protection

BIS-certified 5V 3A power adapter with internal over-voltage and thermal short-circuit protection prevents voltage spikes from entering the printer logic board.

Physical Anti-Theft Chassis

Custom 3D-printed ABS / acrylic enclosure bolts directly to the rear frame of the Epson printer using security torx screws, preventing tampering or unauthorized removal.

7. ITEMIZED BILL OF MATERIALS (BOM) PER STATION						Off-The-Shelf Indian Procurement
#	COMPONENT & TECHNICAL SPECIFICATION	AUTHORIZED INDIAN VENDOR	QTY	UNIT RATE	TOTAL (INR)	
1	Orange Pi Zero 3 (1.5GB LPDDR4) Allwinner H618 Quad-core, Dual-Band Wi-Fi, USB 2.0	Robu.in / Silverline Electronics	1	₹2,200	₹2,200	
2	SanDisk 32GB High-Endurance MicroSD Class 10 A1, rated for continuous IoT write cycles	Amazon India / Local Distributor	1	₹420	₹420	
3	5V 3A USB-C Regulated Power Adapter BIS-certified, surge protected, stable 15W supply	Robu.in / MakerBazar	1	₹350	₹350	
4	1.3" I2C Monochrome OLED Screen 128×64 White display, SH1106 controller, 4-pin I2C	Robu.in / MakerBazar	1	₹260	₹260	
5	Audio-Visual Feedback Subsystem 5V active piezo buzzer, 5mm Green/Blue LEDs, harness	Local Electronics Market	1 set	₹80	₹80	
6	Shielded USB-A to USB-B Cable (0.5m) Short shielded printer data cable for internal mounting	Amazon India	1	₹140	₹140	
7	Laser-Cut Acrylic / 3D ABS Enclosure Direct printer chassis brackets with security fasteners	NERIST 3D Fab Lab / Local Fab	1	₹300	₹300	
8	Thermal Kit & Mounting Hardware Self-adhesive copper heatsinks, brass standoffs, zip ties	Robu.in	1 set	₹100	₹100	
CORE BASE STATION TOTAL (Wi-Fi Enabled Kiosk Module)			1	—	₹3,850	
9	[Optional] Unlocked 4G LTE USB Dongle Independent cellular failover via Jio/Airtel SIM	Amazon India / Flipkart	1	₹1,100	+₹1,100	
10	[Optional] 5V/12V Mini DC UPS Battery Backup Maintains print jobs through hostel load-shedding	Amazon India (Resonate/Oakter)	1	₹850	+₹850	
FULLY LOADED STATION TOTAL (Wi-Fi + 4G SIM Backup + DC UPS)			1	—	₹5,800	

8. PHASED ROLLOUT BUDGET & ENERGY ROI						Financial Feasibility
DEPLOYMENT PHASE	COVERAGE & HOSTEL LOCATIONS	UNITS	BASE CAPEX	LOADED CAPEX	NET SAVINGS VS LAPTOPS	
Phase 1: Pilot Station	Pare Hostel (Block B, Room 29)	1	₹3,850	₹5,800	Save ₹31,150	
Phase 2: Core Quad	Hostel Blocks A, B, C, and D	4	₹15,400	₹23,200	Save ₹1,24,600	
Phase 3: Full Grid	All Hostels (A–H, Girls Hostel, Central)	10	₹38,500	₹58,000	Save ₹3,11,500	

⚡ **Energy Payback ROI:** A laptop draws ~55W continuous (₹3,360/year electricity). The PrintPod draws only ~4W (₹245/year). At an annual power saving of ₹3,115 per unit, each hardware station **fully repays its entire capital cost within 14.8 months purely on electrical savings.**

9. 4-WEEK IMPLEMENTATION ROADMAP			Milestone Schedule
TIMELINE	MILESTONES & ENGINEERING DELIVERABLES	DELIVERABLE VERIFICATION	
Week 1 (Days 1–7)	Procure Orange Pi Zero 3, microSD, OLED & PSU from Robu.in & Amazon India; enclosure CAD drafting.	Component inventory bench testing	
Week 2 (Days 8–14)	Kernel flash (DietPi Bookworm), Read-Only OverlayFS lockdown, CUPS 2.4+ & Epson ESC/P-R setup.	Local USB printer test pass	
Week 3 (Days 15–21)	PrintKurox daemon integration, OLED/piezo status scripts, captive portal watchdog bot deployment.	End-to-end web order test print	
Week 4 (Days 22–28)	Chassis fabrication, 48-hour Pare Hostel stress testing, custodian network IPP setup & handover.	Production deployment sign-off	

10. INSTITUTIONAL ENDORSEMENT & BUDGET SANCTION SHEET		
SUBMITTED BY (DEVELOPER)	RECOMMENDED BY (HOSTEL)	SANCTIONED BY (ADMINISTRATION)
Devananda & Engineering Team Lead System Engineers, PrintKurox NERIST, Nirjuli · Date: _____	Hostel Warden / Custodian Pare Hostel (Block B), NERIST Campus Print Hub · Date: _____	Dean of Student Affairs / TBI Incubation & Student Welfare Authority NERIST, Nirjuli · Date: _____